ChatGPT Prompts for Systematic Thematic Analysis (STA)

Six-step prompts and their rationale, as adopted from Naeem et al. (2025)

These are the ChatGPT prompts used at each of the six steps of the systematic thematic analysis (STA) process introduced by Naeem et al. (2023), together with the rationale (justification) given for each prompt.

These are the Step-by-step prompts, following the order: (1) quotations, (2) keywords, (3) codes, (4) themes, (5) concepts, (6) conceptual model.

Step 1: Familiarisation and Selection of Quotations

Table 1: Prompts to familiarise ChatGPT with the research

Prompt 1A - Prompts to familiarise ChatGPT with the research

This research analyses primary data to achieve the research aim by answering the following research questions. The primary aim of this study is [research purpose]. Based on [theoretical underpinning of research], this study follows [a brief mention of the methodology employed in the study (e.g., constructivist ethnographic approach, exploratory research, inductive approach, qualitative research method)].

Research aim & objectives: [Paste the research aim, objective]

Research question: [Paste the main research question (and optionally information on the research problem].

Context of the research:

- Data type: [Paste the qualitative data from app reviews, semi-structured interviews, focus group discussions, including transcripts]
- Participating and selected Organizations: [Number of participants, background of participants, name of industry, background of selected organisations]

Philosophical underpinnings:

- [Include an explanation of the epistemology by copying and pasting the methodology section of your research]

Theoretical framework:

- Theoretical Framework: [Explain theoretical frameworks used to guide the research]

Data file: [attach the data file to be thematically analysed]

Before reading the manuscript, please review the above background material to familiarise yourself with the research problem, research question and research methodology and context. I will ask you to undertake systematic thematic analysis involving: the selection of quotations, selection of keywords, development of codes, development of themes, and the conceptualisation and development of a conceptual framework using the following six steps to systematic thematic analysis introduced by Naeem et al. (2023). [Ask ChatGPT to write a summary of Naeem et al.'s (2023) systematic thematic analysis, including the 6 Rs to select keywords, the 6 Rs of coding and the 4 Rs of themes]. (ChatGPT prompt supported/developed)

Prompt 1B - Prompts to enable ChatGPT to select quotations from the transcript

Considering the following collective research aim, objectives, methodology, the theoretical framework and philosophical underpinnings, please identify relevant questions to extract excerpts from that data file that answer the research question. The selection process must be guided by the specific context of the study, including its philosophical foundations, research aim, research objectives, research questions, as well as its methodological and theoretical approaches.

- **Research aim & objectives:** [Insert the specific aim and objectives of the research]
- **Methodology:** [Insert the methodology used in the study]
- **Theoretical Framework:** [Insert the theoretical framework]
- **Research Philosophy:** [Insert the philosophical approach guiding the study]
- **Research Question:** [Insert the specific research question(s) the study aims to answer]

Data file: [Paste the data set, or attach a data file, from which quotations will be selected]

Rationale / Justification of the prompt

Naeem et al. (2023, 2024a, 2024b) argued that researchers need to familiarise themselves with the theoretical, contextual and philosophical backgrounds of their research in order to justify their choice of relevant quotations. The need for familiarisation in quotation selection was also suggested by other researchers (Davidson, 2018; Duranti, 2006; Jaffe, 2000; Mondada, 2007; Nowell et al., 2017; Slembrouck, 2007; Thorne, 2000).

Step 2: Selection of Keywords

Table 2: Second step – ChatGPT prompt for selection of keywords.

Prompt 2 - ChatGPT prompt to perform second step

Based on the systematic thematic analysis process introduced by Naeem et al. (2023) and the crucial role of keyword selection in systematic thematic analysis, please consider the following information to aid your selection of appropriate keywords.

Please make sure that the keywords reflect the participants' points of view and insights to directly answer the research question and understand the research phenomenon under study. Please utilise the 6 Rs framework - Realness, Richness, Repetition, Rationale, Repartee and Regal – to guide your selection.

Please consider the following information about the research in your selection of keywords.

Research question: [Insert research question]

Phenomenon under study: [Describe the research aim and research context]

Data file: [Paste the data set or attach the data file]

In selecting keywords, please ensure they:

- are meaningful and help to answer the research question, and
- act as analytical anchors; that is, they help build a narrative based on the participants' own words to answer the research question. (ChatGPT supported/developed prompt).

Rationale / Justification of the prompt

Various authors have identified the following considerations when selecting keywords in thematic analysis.

- Core idea and experience: Select keywords that represent participants' major ideas and capture the major chunks of participants' experiences (Naeem et al., 2023, 2024b).
- Data saturation and integration: Use helpful keywords to integrate various data points and various participant perspectives (Morse, 2015; Naeem et al., 2023).
- Organic selection: Ensure the selected words and phrases are meaningful to build a narrative based on different keywords captured from participant data (Naeem et al., 2023).
- Linking theory and data: Keywords should be related to the research problem, research questions, theories, methodology and research problem (Charma, 2014; Glaser & Strauss, 2009; Naeem et al., 2023; Richards, 2015; Maxwell, 2013).

Step 3: Coding

Table 3: Third step – ChatGPT prompt for Coding.

Prompt 3 - ChatGPT prompt to perform third step

We now move to the third step: Coding. The aim of coding is to use one or more words to jointly represent a group of keywords and quotations to answer the research question. Please take an abductive approach to developing potential meaningful codes, while considering the following information and dataset. These codes should reflect the essence of participants' points of view in a conceptual form. Codes should consist of a maximum of three words.

Please ensure that the suggested codes:

- Contribute to answering the research question and are developed through consideration of keywords and all selected quotations.
- Reflect the 6 Rs coding framework - robust (comprehensive to enable interpretation of the data), reflective (reflect the selected keywords and related quotations), resplendent (rich and conceptually powerful), relevant (related to the research), radical (new contribution), and righteous (ethical and contextually correct).
- Accommodate abductive reasoning by integrating data-driven insights and theoretical frameworks.

Data set segments: [Paste data set segments or specific excerpts]

Identified keywords: [List identified keywords from Step 2]

Selected quotations: [List identified quotations from Step 1]

Research question and aim: [Insert research question and research aim] - (ChatGPT supported/developed prompt)

Rationale / Justification of the prompt

- Saldaña (2013) and Naeem et al. (2023) proposed definitions of coding. Codes should be developed to present the combined meanings of related quotations and selected keywords (Naeem et al., 2023).
- Codes offer new insights to answer the research questions (Saldaña, 2013).
- Coding can be done based on inductive, deductive or abductive reasoning (Attride-Stirling, 2001; Braun & Clarke, 2006; Creswell & Poth, 2018; Elo & Kyngäs, 2008; Tavory & Timmermans, 2014; Thomas, 2006).
- Coding should create robust, reflective, resplendent, relevant, radical and righteous codes, known as the 6 Rs of coding (Naeem et al., 2023).

Step 4: Theming

Table 4: Fourth step – ChatGPT prompt for theming.

Prompt 4 - ChatGPT prompt to perform fourth step

Theming is a process of clustering codes based on their relationship to one another, in that they either fully or partially answer the research question. Given a set of codes and keywords, please suggest appropriate themes to answer the following research question and achieve the research aim. Your theme development should be guided by the 4 Rs framework of theming – Reciprocal (connect the codes in a meaningful way), Recognisable (reflect the actual data), Responsive (responsive to the research question and aim), Resourceful (rich to tell the story of the research).

Make sure the themes are based on the following keywords and codes, and they helpfully answer the research question and aim.

The proposed themes should be different from each other yet interconnected to answer the research aim.

All suggested themes together need to reflect the data appropriately, in terms of theoretical and conceptual adequacy, and comprehensiveness in achieving the research aim.

Use this information to create themes in your mind.

- Coded data segments: [Insert the coded data segments or any other collected data]

- Identified keywords and codes: [paste the list of identified keywords and codes from the previous steps here].
- Research objective and question: [paste the research objective and questions here]
- Theoretical underpinning: [paste the theoretical underpinning here]

(ChatGPT aided/developed prompt) (ChatGPT supported/developed prompt)

Rationale / Justification of the prompt

- Theming is the act of clustering codes based on their relatedness to each other, to answer research question(s), holistically or partially (Naeem et al., 2023; 2024b).
- Themes need to be based within the theoretical and philosophical foundation of the research and must closely link to the context of the research to develop answers to the research questions (Bulmer, 1979; Coffey & Atkinson, 1996; DeSantis & Ugarriza, 2000).
- Consideration of the 4 Rs of theming: Reciprocal, Recognisable, Responsive and Resourceful (Naeem et al., 2023).

Step 5: Conceptualisation

Table 5: Fifth step – ChatGPT prompt for conceptualisation.

Prompt 5 – ChatGPT prompt to perform fifth step

In the conceptualisation step, we aim to develop concepts based on our interpretation of themes, keywords and codes. The concepts should link with theory in the context of this research. Concepts should help answer the research questions. Please define each concept. The definitions should adhere to the principles of clarity and cohere with the theoretical and philosophical context of this research.

To guide this process, please consider the following.

- Research question: [Insert the specific research question]
- Theoretical underpinnings: [Detail the major factors of the theory]
- Themes: [Insert themes from previous step]

Identified keywords and codes: [List the identified keywords and codes from previous steps] (ChatGPT supported/developed prompt)

Rationale / Justification of the prompt

Development of concepts in relation to the data and theoretical underpinning (Babbie, 2016; Byrne, 2015; Kerlinger & Lee, 2000; Naeem et al., 2023, 2024b).

Step 6: Conceptual Model

Table 6: Sixth step – ChatGPT prompt for conceptual model

Prompt 6 – ChatGPT prompt to perform sixth step

In the last step in the structured thematic analysis process described by Naeem et al. (2023), your goal is to synthesise the research findings into a conceptual model. Link these concepts with a figure or diagram that proposes a feasible solution to the following research problem we want to address. The conceptual model must also contribute to managerial practices and the following theory. The proposed conceptual model should answer the following questions. Additionally, you need to keep in mind what the key concepts of existing theory and what are the new concept developed through this research? How are the newly developed concepts related to the already existing theory? What makes these concepts closely related in the context of this research?

As you formulate the conceptual framework, consider the following.

- Research question and objectives: [Insert the specific research question and objectives of the research]

- Theoretical underpinnings: [Detail the theoretical frameworks or underpinnings that guide the study]
- Identified keywords, codes and themes: [List the keywords, codes, and themes identified in previous steps]
- Philosophical underpinnings: [Include an explanation of the epistemology – you can copy and paste the methodology section of your research]
- Theoretical framework: [Explain theoretical frameworks used to guide the research]

The conceptual model must build a story that explains why the findings are meaningful and how they work together to answer the research question. *This content is tailored from OpenAI (ChatGPT supported/developed prompt)

Rationale / Justification of the prompt

The conceptual model should be theoretically and philosophically grounded and evidence-based on actual data (Naeem et al., 2023). These insights derived from a conceptual model should lead to both theoretical and practical understanding of the research problem, which, in turn, help guide decisions (Corley & Gioia, 2011; Smith et al., 2009; Whetten, 1989; Whetten 1989, 1989).

A good conceptual model should explain the how, why and what questions (Wertz et al., 2011).

Reference

Naeem, M., Smith, T., & Thomas, L. (2025). Thematic analysis and artificial intelligence: A step-by-step process for using ChatGPT in thematic analysis. *International Journal of Qualitative Methods.*